

The Study of Blockchain Technology for Copyright Protection and Resource Sharing Models in Library Digital Resources

Tao,Ran

Lingnan Normal University, Zhanjiang, Guangdong, 524048, China

Abstract: The digital transformation of libraries has made the management and protection of digital resources more complex, especially concerning copyright protection and resource sharing. Traditional centralized systems for managing digital rights are increasingly inadequate in addressing the unique challenges of the digital age. Blockchain technology, with its inherent features of decentralization, transparency, and immutability, offers promising solutions to these issues. This paper explores the potential applications of blockchain in library systems, focusing on its ability to safeguard digital copyright and facilitate resource sharing between institutions. By utilizing blockchain's decentralized ledger, libraries can create secure, transparent, and automated systems for managing digital rights, ensuring that authors and copyright holders are fairly compensated. Furthermore, the paper discusses the challenges libraries face in adopting blockchain, including technological scalability, privacy concerns, and legal integration. By examining emerging trends in blockchain and considering the future potential of this technology, the paper highlights how blockchain can transform library operations and the sharing of digital resources. The research suggests that with ongoing advancements, blockchain will play a key role in enhancing the security, efficiency, and transparency of digital resource management in libraries.

Keywords: Blockchain technology; Digital copyright protection; Resource sharing; Library systems; Decentralization

DOI: 10.62639/sspjiss10.20250205

1. Introduction

The digital era has fundamentally altered the way libraries manage and share resources, with digital content now replacing traditional physical media. As libraries expand their digital collections, they face increasingly complex challenges related to copyright protection and resource sharing. Ensuring that digital content is appropriately protected from unauthorized use and facilitating the sharing of resources between institutions in a secure and efficient manner are two key issues that libraries must address. Traditional copyright management systems are often centralized and prone to issues such as fraud, inefficiency, and data manipulation, creating an urgent need for a more robust and transparent solution.

Blockchain technology offers a promising alternative to traditional systems due to its decentralized structure, which makes it difficult for any single entity to control or alter the data recorded on the ledger. With features such as immutability, transparency, and traceability, blockchain can ensure that the rights of digital content creators are protected while allowing for efficient resource sharing between libraries. By utilizing smart contracts, blockchain can also automate processes such as content licensing, copyright transactions, and resource access, thereby reducing administrative burdens and costs.

This paper aims to explore how blockchain technology can be used to address the challenges of digital copyright protection and resource sharing in libraries. It examines the advantages of adopting blockchain, identifies the key challenges that libraries may face in implementation, and suggests strategies to overcome these challenges.

(Manuscript NO.: JISS-25-5-1002)

About the Author

Tao,Ran (1975-), female, Han ethnicity, Jiamusi City, Heilongjiang Province, Bachelor's Degree. Associate Research Librarian. Research interests: Library digital resources, information technology.

Additionally, the paper discusses the future trends in blockchain technology and its potential to revolutionize library systems, making them more efficient, secure, and user-friendly.

2. Overview of Blockchain Technology

Blockchain technology is a distributed ledger system that records transactions across multiple computers in a way that ensures the data cannot be altered or deleted without the consensus of the network participants. Each record, or "block," contains a timestamp and a link to the previous block, creating an immutable chain of information. The decentralized nature of blockchain ensures that no single party controls the data, making it resistant to fraud and manipulation.

In the context of library systems, blockchain can address several key challenges in the management of digital resources. First, it offers a solution for digital copyright protection. Traditional systems of copyright management often rely on centralized databases that can be easily compromised or altered. Blockchain's decentralized ledger ensures that all records related to a particular digital resource—such as ownership, licensing agreements, and usage rights—are securely stored and cannot be tampered with. This provides a transparent and trustworthy record of who owns the copyright to a resource and how it can be used or shared.

Additionally, blockchain can facilitate resource sharing between libraries. Currently, libraries often face difficulties in sharing digital resources due to issues related to copyright compliance, trust, and security. Blockchain technology enables a decentralized and transparent way to track and record transactions between institutions, ensuring that resources are shared in a secure and traceable manner. Through blockchain's use of smart contracts, libraries can automate processes such as resource lending, access control, and payment, reducing administrative overhead and ensuring that copyright holders are compensated fairly.

3. Key Issues in the Application of Blockchain in Libraries

Despite the promising benefits of blockchain technology, several issues need to be addressed before it can be widely implemented in library systems. These challenges primarily relate to technological scalability, privacy concerns, legal integration, and the practical difficulties of adoption.

One of the main technological barriers is scalability. Blockchain systems, particularly those that use Proof of Work (PoW) as a consensus mechanism, can be slow and resource-intensive. For libraries managing large collections of digital resources, scalability is a significant concern, as the blockchain system must be able to process a high volume of transactions efficiently. Solutions such as Proof of Stake (PoS) and other consensus mechanisms that are less resource-intensive may offer a more scalable solution for library applications. Additionally, layer-2 solutions, such as off-chain protocols, can help alleviate scalability issues by processing transactions outside the main blockchain network, thus reducing congestion.

Privacy is another challenge that blockchain faces in library applications. While blockchain's transparency is a key advantage in ensuring the integrity of digital rights, it can conflict with the need for privacy, especially in a setting like a library where users may have sensitive personal information. Privacy-enhancing technologies, such as zero-knowledge proofs, can help address this concern by allowing data verification without revealing the underlying information. However, the implementation of these technologies in blockchain systems is still developing and may require additional research and testing.

Legal and regulatory issues also pose a significant challenge to the adoption of blockchain. The decentralized nature of blockchain can create difficulties in aligning blockchain-based copyright systems with existing legal frameworks, which are typically designed for centralized models. Copyright laws and intellectual property

regulations may need to be adapted or reinterpreted to accommodate blockchain systems, and libraries will need to collaborate with legal professionals to ensure compliance.

Finally, the practical implementation of blockchain technology in libraries is hindered by issues such as integration with existing systems, the cost of adoption, and the need for staff training. Libraries must invest in the necessary infrastructure and resources to support blockchain-based systems, and staff must be educated on how to use these new technologies effectively.

4. Strategies for Overcoming Challenges

One key strategy for addressing scalability is to adopt blockchain systems that use more energy-efficient consensus mechanisms, such as Proof of Stake (PoS) or Delegated Proof of Stake (DPoS). These mechanisms are less resource-intensive and can process transactions more quickly, making them more suitable for library applications that require fast and efficient processing of digital rights transactions. Additionally, using layer-2 solutions, such as payment channels or sidechains, can help improve the scalability of blockchain systems by processing transactions off-chain and only recording final outcomes on the main blockchain.

To address privacy concerns, libraries can implement privacy-enhancing technologies like zero-knowledge proofs, which allow transactions to be verified without revealing sensitive user data. This ensures that the privacy of library users is maintained while still taking advantage of blockchain's transparency and security features. Libraries may also consider hybrid blockchain models, where sensitive data is stored off-chain while only relevant transaction records are stored on the blockchain, thus balancing transparency with privacy.

Legal challenges can be mitigated through collaboration between libraries, blockchain developers, and legal experts. By working together, these stakeholders can develop frameworks that align blockchain-based copyright systems with existing intellectual property laws. In some cases, new regulations may need to be created to address the unique characteristics of decentralized systems. Libraries should also stay informed about developments in blockchain-related legislation to ensure compliance as the legal landscape evolves.

Finally, to overcome the practical challenges of adoption, libraries can focus on training staff and building partnerships with blockchain technology providers. By educating staff on the benefits and implementation of blockchain systems, libraries can ensure smooth integration and reduce resistance to change. Additionally, libraries can collaborate with other institutions to share resources and best practices, making the adoption process more cost-effective and efficient.

5. Trends and Future Prospects

As blockchain technology continues to evolve, several emerging trends are likely to shape its future application in libraries. These trends include the development of cross-chain interoperability, the increasing adoption of Non-Fungible Tokens (NFTs) for copyright management, and the integration of artificial intelligence (AI) with blockchain.

Cross-chain interoperability is becoming increasingly important as multiple blockchain platforms are developed. Cross-chain protocols will allow libraries to share resources and data across different blockchain networks, enabling seamless collaboration between institutions using different blockchain systems. This interoperability could greatly enhance the efficiency of resource sharing between libraries and reduce the need for centralized intermediaries.

Non-Fungible Tokens (NFTs) have gained significant attention in recent years, particularly in the art and

entertainment industries. NFTs offer a unique way to represent ownership and provenance of digital resources, making them ideal for libraries managing rare or high-value digital assets. By associating digital content with NFTs on a blockchain, libraries can ensure that ownership rights are clearly defined and tracked. This could also facilitate the transfer and licensing of digital resources, ensuring that authors and copyright holders are fairly compensated for the use of their work.

The integration of artificial intelligence (AI) with blockchain technology offers exciting possibilities for library systems. AI could be used to automate tasks such as cataloging, user recommendation systems, and content discovery, while blockchain ensures that the ownership and usage rights of digital resources are secure and transparent. By combining AI's ability to analyze and process large volumes of data with blockchain's secure and transparent record-keeping, libraries can create smarter, more efficient systems for managing digital resources.

6. Conclusion

The application of blockchain technology in libraries offers a transformative solution to the challenges of digital copyright protection and resource sharing. Blockchain's decentralized, immutable, and transparent nature ensures that digital resources can be securely tracked and managed, providing a trustworthy system for copyright protection and enabling efficient resource sharing between institutions. Despite challenges such as scalability, privacy concerns, and legal integration, blockchain has the potential to revolutionize library systems and improve the management of digital resources.

To overcome these challenges, libraries can adopt energy-efficient consensus mechanisms, integrate privacy-enhancing technologies, and collaborate with legal experts to ensure compliance with intellectual property laws. Additionally, libraries must invest in staff training and partnerships with technology providers to facilitate the adoption of blockchain solutions.

Emerging trends such as cross-chain interoperability, NFTs, and AI integration offer exciting possibilities for the future of blockchain in libraries. As blockchain technology continues to evolve, it is likely to play an increasingly important role in the management and sharing of digital resources. By embracing blockchain, libraries can enhance the security, transparency, and efficiency of their operations, paving the way for more effective digital resource management and collaboration across institutions.

In conclusion, blockchain technology has the potential to revolutionize the way libraries protect and share digital resources, offering a more secure and efficient solution to the challenges of digital copyright and resource management. As the technology matures, libraries must continue to explore its applications and overcome the challenges of implementation to fully realize its benefits.

References

- [1] Yufang Xiao, Ruifang Xiao et al. "Digital Copyright Management Strategy of Digital Library Based on Blockchain Technology," *Journal of Combinatorial Mathematics and Combinatorial Computing*[2024-03-31].
- [2] R. Xie, Min Tang. "A digital resource copyright protection scheme based on blockchain cross-chain technology." *Heliyon*[2024-09-01].
- [3] A. Savelyev. "Copyright in the Blockchain Era: Promises and Challenges." *ERN: Regulation (IO) (Topic)*[2017-11-21].